Heurion 2.0: A Contract-Guarded Dual-Surface Architecture with Self-Evolving Memory for Citation-Grounded Clinical Research Authoring

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Abstract

General-purpose large language model (LLM) agents typically edit documents via shell scripts (e.g., `python-docx`) and post-hoc file checks. In collaborative clinical research authoring, this workflow risks corrupting journal Office Open XML (`.docx`/`.pptx`) formatting, overwriting concurrent clinician edits, inserting unverified bibliographic identifiers (DOIs/PMIDs), exposing Protected Health Information (PHI) to cloud APIs, and repeating operational guard failures across sessions.

We present **Heurion 2.0**, a clinical authoring platform built on three mechanisms: (1) a *Contract-Guarded Dual-Surface CRDT Engine* (ProseMirror + Yjs) that projects a rich-text editor for clinicians and a guarded Annotated Markdown interface for AI agents, providing pre-write concurrency and citation guards, per-turn selective undo, and original-slice OOXML export; (2) a *Local Two-Stage Neuro-Symbolic PHI Funnel* that pairs sub-millisecond checksum/regex validators with a pre-masked local 1.5B small language model (`qwen2.5:1.5b`) and deterministic biomedical output filtering; and (3) a *Self-Evolving Procedural Memory System* that crystallizes actionable repair rules from runtime guard rejections and clinician turn reverts, isolated from declarative style preferences.

Across 10,769 analysed live LLM turns on 20 open-access PubMed Central (PMC) clinical manuscripts and 1,010 guard evaluation records under pre-specified protocols: (1) in a confirmatory 2,340-turn three-arm manuscript co-editing benchmark across 10 unseen 2026 PMC manuscripts, Heurion 2.0 reduced turn rework to 4.9% (2.1% after excluding evaluator false positives, with 0 lost clinician edits) compared to 14.0% for direct file scripting (OR = 32.47, Holm $p = 1.03 \times 10^{-16}$) and 10.3% for tool-assisted file scripting (OR = 9.66, Holm $p = 1.23 \times 10^{-8}$), while lowering per-turn API cost by 30%–33%; (2) on a locked 100-note Chinese clinical de-identification test set (745 PHI spans) and 200 hard-negative clinical sentences, the two-stage funnel raised PHI recall from 27.4% (regex alone; 3.5% on narrative spans) to 87.0% (83.3% on narrative spans) at 97.6% precision and a 6.0% hard-negative false-positive rate (vs. 92.0% for the unvalidated SLM); and (3) in a confirmatory 4,608-turn, 4-arm longitudinal ablation across 16 users and 8 sessions, online procedural rule learning reduced pre-write guard rejections by 63.7% (RR = 0.363–0.369 vs. no memory, static rules, and length-matched placebo; Holm $p < 10^{-53}$) and raised task completion from 81.1% to 92.8% (100.0% in held-out sessions).

Keywords: Heurion 2.0, Collaborative AI Agents, Clinical Research Authoring, Yjs CRDT Co-Editing, Office Open XML (`.docx`/`.pptx`) Preservation, Model Context Protocol (MCP), Citation Hallucination Prevention, PHI Redaction, Self-Evolving Agent Memory

Statement of Significance

Problem: Autonomous LLM agents that edit clinical manuscripts via file scripts risk corrupting journal OOXML formatting, overwriting concurrent clinician edits, leaking narrative Protected Health Information (PHI) to cloud APIs, and repeating operational guard violations across sessions.

What is Already Known: Deterministic regex scrubbers miss unstructured narrative PHI, whereas unvalidated small language models suffer from severe clinical false positives ($> 90\%$). Static system-prompt rules cannot anticipate evolving collaborator edits or uncovered document constraints.

What This Paper Adds: Heurion 2.0 introduces (i) a contract-guarded Yjs CRDT co-editing engine that cuts manuscript rework from 10.3%–14.0% to 4.9% (OR = 9.66–32.47, Holm $p \leq 1.23 \times 10^{-8}$) and lowers API cost by 30%–33% over script agents; (ii) a local two-stage PHI funnel achieving 87.0% overall (83.3% narrative) recall at 97.6% precision and 6.0% hard-negative FPR; and (iii) an online procedural rule crystallization engine that reduces guard rejections by 63.7% (RR = 0.363–0.369, $p < 10^{-53}$) and achieves 100% held-out task completion across 4,608 longitudinal turns.

Highlights:

- Dual-surface CRDT core preserves OOXML slices and blocks concurrent human-edit loss.
- Confirmatory 2,340-turn benchmark cuts rework (OR = 9.66–32.47) and cost by 30–33%.
- Local 1.5B SLM + symbolic validator reaches 83.3% narrative PHI recall at 6.0% FPR.
- Online procedural rule learning cuts guard rejections by 63.7% across 4,608 turns.
- Four-arm ablation isolates online learning from static rules and prompt placebo.

1 Introduction

Clinical and biomedical research workflows require high standards of traceability, structural fidelity, and regulatory compliance. A single clinical study typically spans patient longitudinal record curation, cohort screening, biostatistical modeling (e.g., baseline Table 1 generation, Kaplan–Meier survival curves, Cox proportional hazards regression, CONSORT flow diagrams), and the iterative authoring of peer-reviewed manuscripts (`.docx`) and academic presentation decks (`.pptx`) [1, 2]. While recent agentic systems have demonstrated strong capabilities in code generation [3, 4] and open-domain literature synthesis [5, 6], deploying autonomous agents into clinical research environments exposes four architectural barriers:

1. **The Free-Form File Mutation Trap:** Conventional script-based document agents treat `.docx` and `.pptx` files as opaque disk artifacts manipulated via generated Python scripts (e.g., `python-docx`) followed by post-hoc quality checks. In practice, script-based rewrites can strip publisher XML attributes, lose paragraph addressing identifiers (`w14:paraId`) during LibreOffice layout passes, suffer from punctuation/whitespace reproduction drift in full-block replacements, and cannot selectively undo an AI turn once a human co-author makes concurrent edits.
2. **Citation Fabrication and Ungrounded Clinical Claims:** LLMs frequently hallucinate plausible-looking Digital Object Identifiers (DOIs), PubMed IDs (PMIDs), or numerical hazard ratios (HR) and p-values [7, 8, 9]. Post-hoc prompting alone cannot guarantee that every inline reference corresponds to a verified bibliographic record or that statistical assertions match the cited source text.
3. **Patient Privacy (PHI) Exposure in Clinical Narratives:** Clinical investigators work with raw electronic medical records containing Protected Health Information (patient names, national ID numbers, admission codes, phone numbers, and residential addresses) [10, 11]. Passing unredacted records to external cloud LLMs conflicts with health-data protection law in both jurisdictions we target: China's Personal Information Protection Law (PIPL) classifies medical and health information as sensitive personal information that may be processed only for a specific purpose, with separate consent and strict protective measures,

and imposes additional conditions on cross-border transfer [12]; the U.S. HIPAA Privacy Rule restricts disclosure of PHI unless it is de-identified, for example by removing the 18 Safe Harbor identifiers [13]. Yet purely deterministic regex scrubbers miss unstructured narrative PHI in ward-round prose, whereas unvalidated small local models frequently misclassify biomedical eponyms and anatomical terms as personal identifiers.

4. **Cross-Session Amnesia and Memory Pollution:** When an agent encounters a structural guard rejection (e.g., attempting to overwrite a human-edited block or inserting an unregistered citation) or when a clinician reverts an AI turn, stateless agents repeat the same operational failure in subsequent turns. Naively appending all historical observations into a single flat memory pool mixes subjective user writing preferences with hard operational constraints, exhausting context budgets with low-utility noise [14, 15].

1.1 Why Script-and-Audit File Mutation Fails in Collaborative Authoring

When general-purpose coding agents operate inside a workspace shell, they typically invoke Python scripts (`python-docx` or XML manipulators) to unpack, mutate, and repackage `.docx` archives on disk, optionally pairing edits with a post-turn audit script that checks text survival rates, paragraph drift, or DOI validity. Collaborative clinical authoring exposes three structural limitations in this "script-and-audit" paradigm:

1. **AI Editing Can Disrupt Document Structure:** When an agent edits a manuscript via ad-hoc scripts or whole-block re-serialization, it risks altering the underlying OOXML structure—stripping journal-template paragraph and run properties (`<w:pPr>`, `<w:rPr>`), breaking list numbering chains and multi-row table merges, detaching inline review comments and track-changes spans, and losing paragraph identifiers (`w14:paraId`) across layout passes.

2. **Risk of Overwriting Concurrent Human Edits:** Because script-based agents read a file snapshot, rewrite whole paragraphs or sections, and write the file back, any concurrent edits made by the clinician mid-turn (such as inserting a clinical caveat in the same paragraph, adjusting a statistic, or attaching a comment) can be overwritten. Reconciling an AI-rewritten file with concurrent browser edits via post-hoc three-way merges is brittle.

3. **Post-Hoc Auditing Cannot Prevent Destructive Writes:** Auditing *after* the write occurs only after mutation: once a script has altered document formatting or overwritten a co-author's edit, a post-hoc check can only flag the discrepancy or trigger a coarse file-level rollback that discards the entire turn. Preventing structural corruption and human-edit loss requires replacing free-form file mutation with a shared, fine-grained collaborative data structure and pre-write structural guards.

1.2 The Heurion 2.0 Architecture and Contributions

Heurion 2.0 shifts from *post-hoc file auditing* to *pre-write structural contracts over a shared CRDT core*, coupled with a *signal-driven self-evolving execution harness*. Our primary contributions are threefold:

- **Contract-Guarded Dual-Surface CRDT Engine (Section 3):** We design a unified ProseMirror + Yjs document core [16, 17] exposed to clinicians via a real-time rich-text editor and to AI agents via Model Context Protocol (MCP) [18] Annotated Markdown primitives, featuring persistent block addressing (`{#id}`), node-level optimistic concurrency control (`base_rev`), minimal-diff XML fragment updates that preserve inline marks and comment anchors, per-turn selective CRDT undo (`revertTurn`), and raw-slice OOXML patch export (`node_src`) that writes back original XML bytes for unmodified blocks.

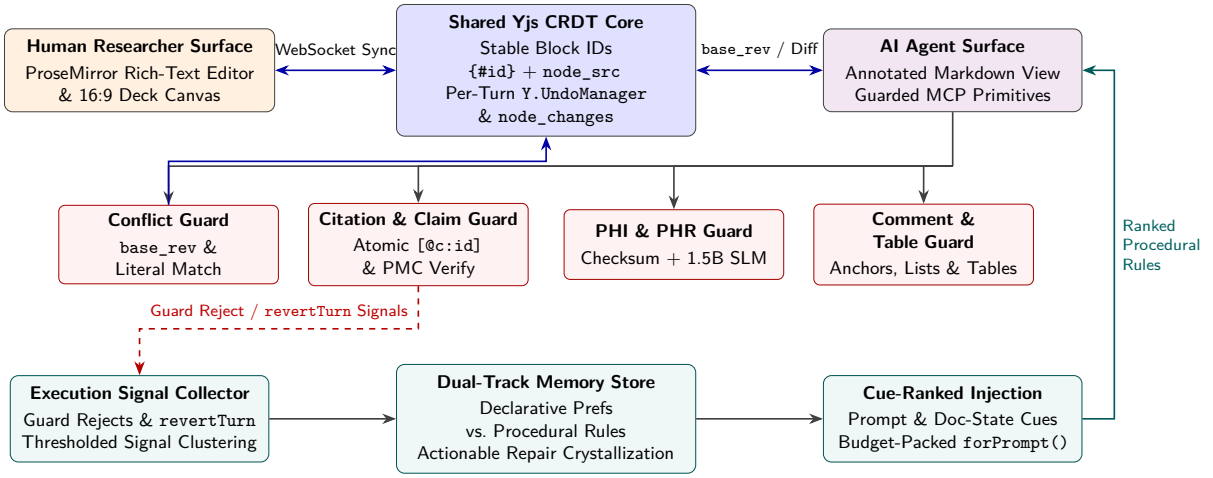


Figure 1: System architecture of Heurion 2.0, a contract-guarded clinical research and scientific authoring platform. Top: Human researchers and AI agents co-edit a shared Yjs CRDT document and slide core through synchronized surface projections (ProseMirror rich-text/deck canvas and Annotated Markdown MCP tools). Middle: All AI mutations pass through deterministic pre-write structural guards (Conflict Guard, Citation & Claim Guard, PHI & PHR Guard, and Comment & Table Guard) before committing to the CRDT state. Bottom: Runtime guard rejections and human turn reverts feed a dual-track self-evolving procedural memory pipeline that crystallizes actionable repair rules and dynamically injects context-relevant directives into future turns.

- **Pre-Write Citation and Two-Stage Neuro-Symbolic PHI Guards (Section 4):** We enforce a syntactic pre-write ban on raw inline DOIs/PMIDs, requiring registration via Crossref, PubMed, and OpenAlex into atomic citation nodes (`[@c:id]`), and design a two-stage PHI redaction cascade that pairs sub-millisecond checksum/regex validators with a pre-masked local 1.5B small language model (`qwen2.5:1.5b`) and deterministic biomedical output filtering to redact both structured identifiers and unstructured clinical narratives within the local loopback boundary. For system completeness, we also outline our multi-agent claim verification pipeline and sandboxed biostatistical tools (which are not quantitatively evaluated in this study).
- **Dual-Track Self-Evolving Procedural Memory (Section 5):** We introduce a self-evolving adaptation loop comprising side-effect-free pre-flight validation probes (`doc_validate` / `dry_run`) and a dual-track memory architecture that physically separates human-approved declarative preferences from runtime-crystallized procedural repair rules ranked by prompt and document-state cues under a fixed context budget.

2 System Overview and Design Principles

Figure 1 illustrates the end-to-end architecture of Heurion 2.0. The platform decouples *structured document/slide state mutation* from *sandboxed statistical computation*.

Heurion 2.0 is built around three core design principles:

1. **Pre-Write Structural Safety:** Every batch of AI edit operations is validated against the current CRDT document state *before* any change is committed. If any operation in a batch targets a block modified by a human since the agent's read revision (`conflict_user_edited`), modifies a block holding an unresolved clinician track-changes suggestion (`pending_suggestion`), fails unique literal substring matching (`text_not_found` or `ambiguous_match`), drops open review-comment anchors without

explicit acknowledgement (`anchor_has_open_comments`), violates list or merged-table structure (`invalid_structure` or `cell_not_found`), or injects unregistered raw citations (`citation_not_registered`), the entire batch is atomically rejected prior to mutating the document, returning the diagnostic code, repair hint, and latest block snapshot.

2. **Local-First Clinical Privacy Boundary:** Raw clinical notes are scrubbed inside the local environment before reaching external LLM APIs: Stage 1 deterministically masks structured and checksummed identifiers in sub-millisecond time, and Stage 2 invokes a local 1.5B small language model on `127.0.0.1` paired with a deterministic biomedical post-validator to mask residual narrative names and addresses without over-redacting clinical terminology.

3. **Declarative–Procedural Memory Separation:** Subjective authoring preferences (declarative memory, e.g., statistical reporting style or spelling conventions) require explicit clinician confirmation in the user interface before activation, whereas objective operational rules for navigating pre-write guards and avoiding turn reverts (procedural memory) evolve automatically from runtime execution signals without polluting declarative memory or requiring manual curation.

3 Dual-Surface Co-Editing and Original-Slice Document Engine

How can autonomous AI agents edit complex Word (`.docx`) manuscripts and PowerPoint (`.pptx`) presentation decks without corrupting journal-template formatting or overwriting concurrent human edits? A central architectural observation in Heurion 2.0 is that Office Open XML (`.docx` and `.pptx`) files are delivery formats, not collaborative editing data structures. When an agent edits a `.docx` file on disk via a Python script while a human views the document in a browser, the two actors operate on divergent state representations. Heurion 2.0 eliminates this divergence by replacing direct file scripting with a Contract-Guarded Dual-Surface CRDT Engine [16, 17].

3.1 Shared CRDT Core with Lifecycle-Stable Block Addressing

In Heurion 2.0, every manuscript and presentation deck resides in a server-authoritative Yjs CRDT document (`Y.Doc`) backed by SQLite state snapshots and an append-only revision log (`doc_commits`). Every block-level ProseMirror node (heading, paragraph, list item, table, figure, or slide element) is assigned a short, lifecycle-stable identifier (`attrs.id`, rendered as `{#id}`). When a human splits a paragraph or pastes content in the browser, the server-side `flush()` handler automatically detects duplicate or missing block IDs and assigns fresh unique IDs inside a system CRDT transaction, ensuring the addressing space remains bijective.

From this single CRDT core, Heurion 2.0 projects two synchronized surfaces:

1. **Human Surface:** A real-time collaborative ProseMirror rich-text editor (for manuscripts) and an interactive 16:9 SVG/HTML slide canvas (for decks) synchronized via WebSocket (`y-protocols/sync`).

2. **Agent Surface:** A compact *Annotated Markdown* view generated on demand via `doc_read` and `slide_read`. Each block is suffixed with its stable address `{#id}`, inline citations are rendered as atomic tokens `[@c:cite_id]`, figures as `![alt](asset:id)`, and active inline review comments as `{#c:comment_id}`.

3.2 Node-Level Conflict Guards and Minimal-Diff Fragment Updates

To prevent an AI turn in Heurion 2.0 from overwriting concurrent human edits, every `doc_edit` call specifies the `base_rev` read by the agent. The platform queries the `node_changes` index for all blocks modified by human actors since `base_rev`:

- **Block-Level Structural Operations (`replace_block`, `delete`):** If a target block has been modified by a human since `base_rev`, the operation is rejected with error code `conflict_user_edited`, returning the human's latest block markdown so the agent can adapt. Similarly, blocks carrying unresolved track-changes suggestions are protected by the `pending_suggestion` guard, and blocks holding open review comments require either preserving the comment anchors or explicitly declaring `drop_anchors` (`anchor_has_open_comments`).
- **Substring Replacement (`replace_text`):** For surgical phrase edits within a paragraph, the literal `find` substring acts as its own optimistic concurrency guard: if the human edited a different sentence in the same paragraph, `find` still matches uniquely and succeeds; if the human altered the target phrase itself or if `find` matches multiple occurrences, validation fails cleanly with `text_not_found` or `ambiguous_match`.
- **Mark-Preserving Minimal Diffs:** Notably, `replace_text` does not re-parse the paragraph from plain text (which would destroy inline bold/italic spans, `[@c:id]` citation nodes, and comment highlights outside the matched range). Instead, it locates the matched character span in the ProseMirror node's flattened text stream and replaces only that slice, after which `writeFragment` updates only the modified XML children between the common prefix and suffix in the underlying `Y.XmlFragment`.

3.3 Per-Turn Selective Undo in CRDT Space (`revertTurn`)

In collaborative scientific writing, a researcher may ask the AI to revise three sections, continue manually editing Section 2, and subsequently decide to undo the AI's changes. Global linear undo (`Ctrl+Z` or version rollback) would destroy the researcher's manual edits in Section 2.

Heurion 2.0 solves this via a two-tier *Per-Turn Selective Undo* mechanism (`revertTurn`):

1. **In-Memory CRDT Origin Tracking:** When an AI turn commits changes to a document, the runtime tags every CRDT transaction with the turn's unique origin identifier and attaches a turn-scoped `Y.UndoManager` tracking only that origin. Invoking `revertTurn(docId, turnId)` executes `undo()` solely on the CRDT items inserted or deleted during that AI turn, leaving concurrent human character insertions—even within the same paragraph—intact.
2. **Persistent Node-Level Fallback across Server Restarts:** Before a turn's first write, the runtime persists the pre-turn Yjs state vector in `turn_doc_bases`. If the server restarts before `revertTurn` is invoked, the engine reconstructs the pre-turn document tree, computes the per-block diff against the turn's recorded `node_changes`, reverts all AI-modified blocks untouched by the user since the turn began, and safely skips blocks subsequently edited by the user (returning their IDs in `skipped`).

3.4 Original-Slice OOXML Patch Export

Clinical manuscripts are frequently imported from journal-specific Word templates containing custom styles, headers/footers, macros, and embedded vector objects. During `.docx` or `.pptx` import, Heurion 2.0 stores the original archive in `doc_packages` and records the raw XML slice (`<w:p>` or `<w:tbl>`) of every imported block in `node_src` keyed by block ID. Upon export (`docx-export.ts`, `pptx-export.ts`), any block whose ProseMirror node is structurally equal (`node.eq(baselineNode)`, meaning its text, inline formatting marks, and attached comment range markers are unchanged) to its imported baseline is written back using its exact original `node_src` XML bytes; only blocks whose text or inline comment/mark structure was mutated during the session are re-serialized, preserving paragraph properties (`<w:pPr>`), run fonts, and paragraph identifiers (`w14:paraId`) from the original template.

4 Citation-Grounded Evidence Loop and Privacy-Preserving Clinical Pipeline

How can a clinical authoring platform prevent LLM citation fabrication while protecting patient privacy (PHI) in medical records? Heurion 2.0 addresses both requirements through deterministic pre-write syntactic guards and a two-stage neuro-symbolic PHI redaction funnel.

4.1 Syntactic Citation Guard and Atomic Citation Nodes

To prevent fabricated references from entering manuscript text, Heurion 2.0 blocks agents from writing raw bibliographic identifiers or hand-numbered brackets in prose. During pre-write validation in `OpService`, every candidate text string is scanned against structural regular expressions detecting bare DOIs (`10.\d{4,9}/…`), bare PMIDs (`PMID:\s*\d+`), or unlinked reference markers. Any violation triggers `citation_not_registered` before modifying the document.

Instead, the agent must invoke `insert_citation`, which queries live metadata APIs (Crossref, NCBI PubMed E-utilities, and OpenAlex), verifies the record's existence, stores the canonical metadata (authors, title, journal, year, volume, pages, DOI/PMID) in the document's `citations` table, and returns a registered token `[@c:cite_id]`. In ProseMirror, `[@c:cite_id]` is parsed into an atomic inline node (`atom: true`) that dynamically numbers itself according to citation order and automatically compiles the terminal References section.

4.2 System Context: Claim Verification and Biostatistical Sandbox

In addition to the pre-write citation guard evaluated in Section 6, the platform includes a multi-agent claim verification service (`ClaimService.verify`) and a sandboxed biostatistical execution toolchain (included here for architectural context; clinician-graded evaluation of these modules is left to future work). Briefly, `verify_claims` segments manuscript blocks into sentences, retrieves PubMed abstracts and available PubMed Central (PMC) full texts for cited `[@c:cite_id]` records, and classifies quantitative assertions into four verdicts (`supported`, `contradicted`, `unsupported`, or `uncited_stat`), attaching problematic claims as inline ProseMirror review comments. For statistical analysis, computation is delegated to isolated Python/R tools (`biostatistician` subagent) rather than free-text LLM arithmetic.

4.3 Multi-Tenant PHI Redaction Funnel

To bridge clinical data management and manuscript authoring without exposing patient identities [10, 11, 19], Heurion 2.0 enforces four privacy safeguards:

- **Per-Tenant Database Isolation & Envelope Encryption:** Each organization tenant operates on a physically isolated SQLite database file (`tenants/<org_id>/platform.db`) with per-tenant Data Encryption Keys (DEKs) wrapped by a master Key Encryption Key (KEK).
- **Two-Stage Neuro-Symbolic PHI Funnel (`phi-scan.ts`):** Using an external cloud LLM to inspect raw clinical text for PHI creates a privacy paradox: sending unredacted patient records to a third-party API in order to redact them already violates tenant data-residency boundaries. Conversely, running a frontier-scale 70B+ parameter model on-premise imposes high GPU requirements for clinical research teams. Heurion 2.0 addresses this trade-off through a task-decomposed, two-stage neuro-symbolic funnel that operates within the local environment:
  1. *Stage 1 — Sub-Millisecond Deterministic Checksum & Subtype Rules (`scanPhi`/`redactPhi`):* Deterministic rules execute first (0.011 ms per note) to validate and mask

mathematically structured or field-labeled identifiers: 18-digit national IDs (ISO 7064 MOD 11-2 checksum + calendar bounds), settlement/bank cards (ISO/IEC 7812 industry prefix + Luhn mod-10 checksum with DOI/URL boundary exclusion), 11-digit mobile numbers, medical record numbers, and co-occurring patient-name + clinical diagnosis pairs spanning both pure-CJK and mixed alphanumeric/Roman-numeral subtypes (e.g., Type-2 diabetic nephropathy, diffuse large B-cell lymphoma).

2. *Stage 2 — Pre-Masked Local Small Language Model with Symbolic Output Guard (`createLocalSlmPhiDetector` + `redactPhiCascaded`):* Because Stage 1 pre-masks structured identifiers before Stage 2 is invoked, the Stage-2 model does not receive raw national IDs, phone numbers, or bank cards, and does not need to perform arithmetic checksum verification. Instead, a compact 1.5B-parameter local small language model (`qwen2.5:1.5b`, 986 MB, served via local Ollama and gated by `assertLocalPhiEndpoint` to allow only loopback `127.0.0.1`/`::1` or RFC 1918 private intranet hosts) extracts open-vocabulary residual spans in free-form ward-round narratives: unlabeled person/relative names (`unlabeled_person_name`) and specific residential/workplace addresses (`address_or_org`). Finally, a deterministic post-validator (`normalizeAndValidateStage2Entity`) verifies exact substring presence in the source text, strips kinship/location prepositions, checks Chinese family-surname membership and medical-suffix exclusions for names, and requires spatial/organizational markers for addresses—filtering out small-model hallucinations and clinical-term false positives. The model thus acts as a candidate generator and the post-validator decides what is masked; text containing no CJK characters (e.g., English manuscript prose) is not sent to the model.

- **Pseudonymous Subject Mapping & PHR Red-Lines:** Within research cohorts, real patient identifiers are replaced by pseudonymous subject codes (`P001`, `P002`, …) via an in-memory translation table. Furthermore, a dedicated Personal Health Record guard (`phr_guard`) enforces read-only access on patient-uploaded primary medical records, preventing AI tools from altering ground-truth clinical evidence.
- **Regulatory Scope:** These mechanisms reduce exposure but do not by themselves establish compliance with either PIPL or HIPAA. Pseudonymous subject codes are de-identification rather than anonymization: under PIPL, data that can be re-linked with a retained mapping table remains personal information [12]. The current scanner implements mainland Chinese identifier formats (resident ID numbers, mobile numbers, hospital record numbers, Chinese surnames/addresses); it does not yet cover HIPAA Safe Harbor identifiers such as Social Security numbers, U.S.-format telephone numbers, full dates, or English geographic subdivisions [13], so U.S. deployment requires extending it.

5 Dual-Track Self-Evolving Procedural Memory

A recurring challenge in contract-guarded agent environments is helping the LLM internalize operational boundaries without manual prompt engineering. When an agent's tool call is blocked by a pre-write guard or reverted by a user, how does the system help the agent adapt both within the current turn and across future authoring sessions?

As shown in the bottom tier of Figure 1, Heurion 2.0 combines intra-turn pre-flight validation probes with a cross-session dual-track procedural memory pipeline:

5.1 Intra-Turn Pre-Flight Probe (`doc_validate` & `dry_run`)

Within a single turn, agents in Heurion 2.0 can invoke `doc_validate` (or pass `dry_run: true` to `doc_edit`) prior to committing complex multi-block edits. The probe executes the full suite

of conflict guards, citation registration checks, anchor integrity checks, and post-edit structural diffs against an ephemeral in-memory clone of the ProseMirror document tree without incrementing `rev`, mutating the Yjs CRDT state, or polluting the undo stack. If a violation is detected, the probe returns `{valid: false, code, hint, current}`, enabling immediate self-correction within the same reasoning turn.

5.2 Dual-Track Memory Separation and Actionable Rule Crystallization

Across sessions, Heurion 2.0 prevents memory pollution by physically separating declarative preferences from procedural rules:

- **Track A — Declarative Memory (`MemoryService`):** Captures domain conventions and author preferences (e.g., "Always report continuous skewed variables as median (IQR)", "Use British English spelling"). Because stylistic preferences are subjective, newly extracted items enter a `candidate` state and require explicit human approval in the UI before becoming `confirmed`.
- **Track B — Procedural Harness Memory (`ProceduralHarnessService`):** Captures objective operational rules for navigating guards, tools, and specialist subagents within each user's isolated namespace (`ownerId`). Whenever `OpService` rejects an AI edit or a user invokes `revertTurn`, the `ExecutionSignals` collector automatically logs a structured runtime signal (recording the document ID, turn ID, signal kind, guard code, and diagnostic detail) into the `execution_signals` table.

At the end of an authoring session, `evolveFromSignals()` groups unconsumed execution signals by signal kind and guard code. Any failure pattern occurring at least twice either increments the evidence counter of an existing procedural rule or crystallizes a new active procedural directive via guard-specific actionable templates (`LEARNED_RULE_TEMPLATES`)—requiring zero manual UI intervention. Rather than emitting generic warnings (e.g., "call `doc_validate` before editing"), each crystallized rule encodes the exact positive parameter or tool-sequencing fix required by the underlying contract (e.g., including leading clause context or explicit `occurrence: 2` for `ambiguous_match`; stripping HTML `<sup>`/`<sub>` tags from `find` while retaining them in `replace` for `text_not_found`; prepending `{op: "move"}` before `{op: "merge"}` in the same batch for `invalid_structure`; calling `comments_list` after `doc_read` and passing `ack_comments` + `comment_reply` for `anchor_has_open_comments`; targeting only the top-left origin `{row: 0, col: 0}` of merged table headers for `cell_not_found`; and restricting edits to the single named block after `user_reverted_turn`).

5.3 Context-Aware Cue Ranking and Prompt Budget Packing

Injecting all procedural rules into every turn would waste context tokens. Before each turn, `ProceduralHarnessService.forPrompt()` scores each active rule in the user's store by combining its historical evidence count with two context cues (since guard codes themselves never appear in a user's natural-language request): (1) a *message cue* matching task keywords in the user's prompt (e.g., DOI/PMID or citation terms for `citation_not_registered`, replacement/statistic verbs for `ambiguous_match` and `text_not_found`, paragraph-merge or transition verbs for `invalid_structure` and `user_reverted_turn`), and (2) a *document-state cue* inspected from the current document (human edits within the last two revisions for `conflict_user_edited`, open comment threads for `anchor_has_open_comments`, pending suggestions for `pending_suggestion`, or the presence of a table for `cell_not_found` and `invalid_table`). Specifically, each rule receives 0.1 points per historical occurrence, +5 points when its message cue matches the user prompt, and +8 points when its document-state cue is active in the target manuscript (heuristic weights set prior to evaluation). Rules are sorted by

score and greedily packed up to at most 4 rules within a 1,000-character budget. To isolate rule content from prompt-length effects in controlled experiments, the service also supports a `placebo` mode (`PROCEDURAL_MODE=placebo`) that appends a length-matched block of generic platform background text alongside the static baseline rules. In multi-tenant deployments, an optional background service (`HarnessEvolutionService`) can also periodically distill high-frequency rules across tenants into a shared profile overlay (`evolved.cordis.yml`).

6 Empirical Evaluation

We evaluate Heurion 2.0 through a two-tier, two-generation empirical protocol (`ANALYSIS_PLAN.md` for v1 and `ANALYSIS_PLAN_v2.md` for v2) combining:

1. **Deterministic and Local-SLM Benchmarks (`run_eval.ts` & `v2/phi/run.ts`):** 1,010 evaluation records against the platform's TypeScript/SQLite/Yjs engine and local Ollama runtime, comprising 50 CRDT concurrency/revert trials (`exp1`), 660 initial citation and template PHI guard records (`exp2b`), and a locked 300-record multi-specialty Chinese clinical PHI de-identification test split (`v2-phi`: 100 diverse clinical notes containing 745 PHI spans [485 narrative, 260 labeled] and 200 hard-negative clinical sentences evaluated across 4 ablation arms). Local Stage-2 PHI extraction used `qwen2.5:1.5b` (986 MB, Ollama v0.20.5 on `http://127.0.0.1:11434/api/chat`, `temperature=0`, `format='json'`).
2. **Pre-Specified Live LLM Evaluations (`ANALYSIS_PLAN.md` & `ANALYSIS_PLAN_v2.md`):** 10,769 analysed live agent turns across 20 open-access (CC BY) clinical research manuscripts from PubMed Central (PMC), comprising: (i) 1,241 v1 baseline turns (162 in `exp2a-papers` and 1,079 in `exp3`); (ii) 4,488 three-arm collaborative manuscript editing turns in `v2-q1` (2,148 development turns in Phase 1a on 10 PMC manuscripts and 2,340 confirmatory turns in Phase 1b on 10 newly selected 2026 PMC manuscripts across stress, multi-turn session, and routine tasks); and (iii) 5,040 longitudinal procedural memory turns in `v2-exp3` (432 pilot turns across 3 arms and 4,608 confirmatory turns across 4 arms $\times$ 16 users $\times$ 8 sessions $\times$ 9 uncovered constraint tasks, with 0 failed or excluded turns). All online turns were executed using model ID `deepseek-flash` via `@deepseek-ai/dsh-sdk-client` against `https://api.deepseek.com/anthropic/v1`. Across longitudinal memory experiments, each turn executes in a fresh agent session and each simulated user evolves within an isolated per-user procedural memory namespace (`ownerId`) without cross-user rule sharing, ensuring that all user trajectories remain statistically independent.

6.1 Experiment 1: Dual-Surface CRDT Engine vs. Script-Based File Mutation

Table 1 contrasts the architectural mechanisms of direct script mutation (`python-docx`), script mutation paired with post-hoc file auditing, and Heurion 2.0.

Experiment 1a: Deterministic Engine Micro-Benchmarks (`exp1`). Table 2 reports in-process micro-benchmarks of Heurion 2.0 over $N = 50$ trials on a six-block `.docx` clinical protocol template (one styled heading, a bold paragraph, an italic paragraph, two list items, and a plain paragraph). In each trial, an AI turn applies two `replace_text` edits (p_1, p_2), a simulated human edits a different phrase inside p_2, a stale `replace_block` is probed, and the AI turn is selectively reverted via `revertTurn`. All 50/50 trials reproduced the original `<w:p>` XML bytes of the untouched heading and final paragraph verbatim via `node_src`, rejected the stale overwrite with `conflict_user_edited`, and undid the AI edits while preserving the human's intra-paragraph edit in p_2 at sub-millisecond latency (0.168 ms validation, 0.338 ms edit, 0.129 ms revert).

Table 1: Architectural mechanism comparison across direct script mutation (`python-docx`), script mutation with post-hoc file auditing, and Heurion 2.0's contract-guarded dual-surface CRDT architecture. Entries describe system design mechanisms, not measured rates.

Dimension	Direct Script (`python-docx`)	Script + Post-Hoc Audit	Heurion 2.0 (Ours)
Source of Truth	Raw `.docx` on Disk	Raw `.docx` + Derived Projection	**Shared Yjs CRDT + SQLite**
Block Addressing	None (Regex/XPath)	File-Native `w14:paraId` / Shape ID	**Server-Assigned `{#id}`**
Export of Untouched Blocks	Full Re-serialization	Whole-File Save	**Original XML Slice (`node_src`)**
Concurrent Human Edits	Silent Overwrite	Post-Hoc Node-Level 3-Way Merge	**Pre-Write `base_rev` Guard**
Undo of One AI Turn	Unsupported	File-Level Rollback	**Per-Turn `Y.UndoManager`**
Unregistered Citations	Not Checked	Post-Write Audit	**Pre-Write Block**
Pre-Flight Validation	None	Subprocess Audit Script	**In-Memory Dry Run**

Table 2: Heurion 2.0 dual-surface CRDT engine micro-benchmark results (`exp1`) over $N = 50$ trials on the synthetic clinical protocol template. Latencies are wall-clock means of in-process calls on a single developer machine.

Metric	Result
Untouched blocks reproduced verbatim on export (heading + final paragraph)	50/50
Stale `replace_block` rejected with `conflict_user_edited`	50/50
Turn revert undoes AI edits while keeping the human edit in the same paragraph	50/50
Mean pre-flight validation latency (`OpService.validate`)	0.168 ms
Mean guarded edit latency (`OpService.edit`, two ops)	0.338 ms
Mean per-turn revert latency (`revertTurn`)	0.129 ms

Experiment 1b: End-to-End Clinical Manuscript Editing—Heurion 2.0 MCP vs. `python-docx` (`exp2a-papers`). To test how the two editing surfaces behave under live LLM tool use on full-length journal manuscripts, we compared Heurion 2.0's MCP surface (`PROCEDURAL_MODE=off`) against a baseline coding agent using the same model (`deepseek-flash`) and identical authoring guidelines (`baseline.cordis.yml`) to edit `manuscript.docx` directly in a workspace directory. In the baseline arm, the agent runs inside the same `DeepSeekHarness` runtime with unrestricted `bash` and `python3` access (including `python-docx`, `zipfile`, and `xml.etree.ElementTree`) and is explicitly instructed to *"preserve the existing formatting, comments and tracked changes"*. Both arms executed 3 paired simulated user seeds (seeds 201–203) × 3 PMC clinical manuscripts × 9 authoring task archetypes (`concurrent_edit`, `citation`, `pending_suggestion`, `open_comment_anchor`, `term_unify`, `table_row`, `out_of_scope_revert`, `p_value_format`, and `move_subsection`), yielding 81 turns per arm (162 turns total; 32,995 untouched paragraphs per arm). Both arms exported standard `.docx` archives evaluated by the same OOXML inspector (`manuscripts/compare.ts`).

Table 3 summarizes three empirical observations from `exp2a-papers`:

1. **Comparable Task Completion and Structural Preservation Rates:** Task completion did not differ significantly between Heurion 2.0 MCP (76/81 = 93.8%) and direct `python-docx` scripting (77/81 = 95.1%, Fisher $p = 1.0000$). Across 32,995 non-target paragraphs, Heurion 2.0 produced 0 unintended text modifications compared to 4 altered non-target paragraphs in the `python-docx` arm ($p = 0.1250$), and retained `w14:paraId` on 41/45 (91.1%) vs. 37/45 (82.2%, $p = 0.3529$) edited paragraphs; neither difference reached statistical significance at $n = 81$ turns per arm. Why were 7 of 32,995 untouched paragraphs in Heurion 2.0 not byte-identical (32,988/32,995 = 99.98% vs. 32,991/32,995 = 99.99%)? Inspection of the turn logs showed that all 7 paragraphs occurred in two `open_comment_anchor` turns (`u3-s3` on `PMC8861838`, 4 paragraphs; `u1-s3` on `PMC10260499`, 3 paragraphs) where the MCP agent voluntarily invoked the `fact_checker` subagent (`delegate_task`), which attached unrequested `<w:commentRangeStart/>` verification comments to those 7 paragraphs

Table 3: End-to-end clinical manuscript editing comparison (`exp2a-papers`, 162 live LLM turns across 81 paired tasks per arm, 3 simulated user seeds × 3 manuscripts × 9 archetypes) between Heurion 2.0 MCP primitives and direct `python-docx` file scripting. 95% confidence intervals use the Wilson score method; p-values are two-sided Fisher's exact tests. †In 2 `open_comment_anchor` turns (`u3-s3`, `u1-s3`), the MCP agent invoked `delegate_task` (`fact_checker`) without being asked to, which attached 7 unrequested verification comments (`<w:commentRangeStart/>`) to non-target paragraphs; their text was unchanged (`content_changed = 0`), but these are AI-initiated writes outside the task scope. ‡In full-paragraph rewrites, the MCP agent passed `drop_anchors: true` (6/9 cases) to clear obsolete range tags after replying to all 9/9 review threads via `comment_reply`, whereas `python-docx` run-level edits left the sibling `<w:commentRangeStart/>` tags in place without replying to `word/comments.xml`.

Evaluation Endpoint	Heurion 2.0 MCP ($n = 81$)	Direct `python-docx` ($n = 81$)	Fisher p
Task completion rate across all 9 archetypes	76/81 (93.8%, CI 86.4–97.3%)	77/81 (95.1%, CI 88.0–98.1%)	1.0000
Untouched paragraphs with unintended content changes	0/32,995 (0.00%)	4/32,995 (0.01%)	0.1250
Untouched paragraphs preserved byte-identically†	32,988/32,995 (99.98%)	32,991/32,995 (99.99%)	0.5488
Edited paragraphs preserving native `w14:paraId`	41/45 (91.1%, CI 79.3–96.5%)	37/45 (82.2%, CI 68.7–90.7%)	0.3529
Edited paragraphs preserving layout, fonts, and emphasis	44/45 (97.8%)	45/45 (100.0%)	1.0000
Review comments read & replied in thread (`acknowledged`)	**9/9 (100.0%, CI 70.1–100%)**	0/9 (0.0%, CI 0.0–29.9%)	**< 0.0001**
Inline `<w:commentRangeStart/>` tag retained (6/9 cleared via `drop_anchors`)‡	3/9 (33.3%, CI 12.1–64.6%)	**9/9 (100.0%, CI 70.1–100%)**	**0.0090**
Clinician pending track-change suggestion preserved	9/9 (100.0%, CI 70.1–100%)	9/9 (100.0%, CI 70.1–100%)	1.0000
Unregistered bare DOI / PMID injected into manuscript body	0/81 (0.0%, CI 0.0–4.5%)	1/81 (1.2%, CI 0.2–6.7%)	1.0000

while leaving their text unchanged (`content_changed = 0`). We therefore report these as out-of-scope side effects: Heurion 2.0 made 0 unintended text changes but 7 unrequested comment insertions on non-target paragraphs.

2. **Comment-Thread Resolution vs. Unassisted File Scripting:** In the `open_comment_anchor` task (which instructs the agent to rewrite a Methods paragraph according to two attached clinician review comments), although the baseline agent had full Python/XML access to `word/comments.xml`, standard `python-docx` scripts written by the agent iterated over `Document.paragraphs` (which does not expose `word/comments.xml`) and lacked a dedicated comment-thread tool or pre-write guard; consequently, the file agent modified the paragraph's text runs while leaving the sibling `<w:commentRangeStart/>` XML tags in place (9/9 vs. 3/9, $p = 0.0090$) and never appended a reply to `word/comments.xml` (0/9 acknowledged). Conversely, Heurion 2.0's `anchor_has_open_comments` guard and `comments_list`/`comment_reply` tools prompted the MCP agent to inspect the comments, declare `drop_anchors` when replacing the full paragraph (6/9 cases), and post structured replies in all 9/9 (100%, $p < 0.0001$) review threads. These replies live in the platform's comment threads: the exported `.docx` does not carry them, and in 4/9 exported files the two original clinician comments were removed along with their anchors, so a reviewer working only in Word would not see how the comments were resolved.

3. **Methodological Note on File-Arm Concurrent Edits:** Both arms preserved 9/9 clinician insertions in `concurrent_edit`. Inspection of the event logs revealed that because the file agent executes a single atomic `bash` call (`Document('manuscript.docx') ... save()`) whereas the harness callback `humanEditsFile` fires upon `tool_result` completion, the simulated clinician insertion in the file arm occurred between separate `bash` invocations rather than mid-flight between an in-memory read and write. Mid-flight stale-snapshot overwrites are evaluated deterministically in Table 2 (50/50 intercepted) and in Experiment 1c below.

Experiment 1c: Confirmatory Three-Arm Collaborative Manuscript Editing (`v2-q1`, 4,488 Turns). Because the initial 162-turn `exp2a-papers` comparison used explicit preservation prompts, single-turn tasks, and a file baseline without literature search tools, we designed a two-phase development–validation benchmark (`ANALYSIS_PLAN_v2.md` §2–§4b) comparing three

Table 4: Confirmatory Phase 1b three-arm clinical manuscript editing results (`v2-q1`, $N = 2,340$ live LLM turns across 10 unseen 2026 PMC manuscripts; 780 paired turns per arm). Primary accuracy models use mixed-effects logistic regression on stress + multi-turn session turns ($n = 699$/arm); primary cost models use linear mixed-effects regression on log-cost across all 780 turns/arm, with Holm step-down correction across the four pre-specified confirmatory tests.

Endpoint / Stratum	Heurion 2.0 (P)	Direct Script (B1)	Assisted Script (B2)	B1 vs. P Effect (Holm p)	B2 vs. P Effect (Holm p)
Stress Tasks Rework ($n = 123$/arm)	**2.4**% (3/123)	20.3% (25/123)	9.8% (12/123)	—	—
Multi-Turn Session Rework ($n = 576$/arm)	**5.4**% (31/576)	13.4% (77/576)	10.6% (61/576)	—	—
Routine Tasks Rework ($n = 81$/arm)	**4.9**% (4/81)	8.6% (7/81)	8.6% (7/81)	—	—
All Turns Rework ($n = 780$/arm)	**4.9**% (**38/780**)	14.0% (109/780)	10.3% (80/780)	OR = **32.47** (**1.03×10^{-16}**)	OR = **9.66** (**1.23×10^{-8}**)
Severe Rework ($n = 780$/arm)	**1.2**% (9/780)	4.4% (34/780)	5.4% (42/780)	OR = 27.98 ($p = 4.20 \times 10^{-6}$)	OR = 51.95 ($p = 7.87 \times 10^{-8}$)
Lost Human/Collaborator Content	**0**	18	22	—	—
OOXML File Structural Corruption	**1**	32	3	—	—
Out-of-Scope Collateral Edits	**1**	9	8	—	—
Fabricated / Mismatched Citations	**0**	1	3	—	—
Total API Cost (780 turns/arm)	**$9.26**	$13.21	$13.05	GMR = **0.70** (**3.73×10^{-73}**)	GMR = **0.67** (**2.99×10^{-95}**)
Median Turn Latency (Stress / Session)	**33.5** s/**15.5** s	57.3 s/26.5 s	61.6 s/27.9 s	−41.5% / −41.5%	−45.6% / −44.4%

arms under natural clinician instructions, true mid-turn/inter-turn concurrent human edits, complex journal OOXML templates (188/188 fidelity checks passing in `v2/fidelity/run.ts`), and identical literature search access:

- **Arm P (Heurion 2.0 Platform):** Dual-surface CRDT core with document-scoped MCP tools (63 tools exposed per manuscript via `mcp/scope.ts`) and pre-write structural/citation guards.
- **Arm B1 (Direct File Scripting):** Coding agent with `bash`/`python3` (`python-docx`, `lxml`) editing `manuscript.docx`.
- **Arm B2 (Tool-Assisted File Scripting + Audit):** Same as `B1` plus helper inspection/audit scripts and the same live Crossref/PubMed/OpenAlex literature search tools as `P`.

In the **Development Phase (1a, 2,148 turns across 10 PMC manuscripts)**, `P` eliminated lost concurrent clinician sentences (0 vs. 23 in `B1` and 17 in `B2`) and OOXML structural corruption (0 vs. 32 in `B1` and 3 in `B2`), achieving a rework odds ratio of OR = 3.62 vs. `B1` (Holm $p = 2.14 \times 10^{-6}$) and 20% lower per-turn cost (GMR = 0.80, $p < 10^{-21}$), while exposing five platform gaps (unprotected platform-native user suggestions, lack of scoped batch text replacement, unverified citation claim support at insertion time, manual paragraph merges dropping inline marks, and stripped citation subtitles). After implementing fixes `F1`–`F5` and freezing tag `paper-eval-v2-p1b`, we executed the **Confirmatory Phase (1b, 2,340 turns = 780 turns/arm across 10 newly selected 2026 CC BY PMC manuscripts: 123 single-turn stress tasks, 576 multi-turn collaborative session turns [9 papers × 8 sessions × 8 turns], and 81 routine tasks per arm)**.

As shown in Table 4, all four pre-specified confirmatory hypotheses in Phase 1b passed after Holm correction: Heurion 2.0 reduced rework odds by OR = 32.47 (95% CI 14.39–73.26, Holm $p = 1.03 \times 10^{-16}$) against `B1` and OR = 9.66 (95% CI 4.43–21.08, Holm $p = 1.23 \times 10^{-8}$) against `B2`, while reducing API cost by 30% (GMR = 0.70, 95% CI 0.67–0.73, Holm $p = 3.73 \times 10^{-73}$) and 33% (GMR = 0.67, 95% CI 0.64–0.69, Holm $p = 2.99 \times 10^{-95}$) and cutting median turn latency by 41%–46%. Pre-write guards intercepted 138 unsafe operations across 111 platform turns, preventing any loss of concurrent human edits (0 vs. 18 and 22). Post-hoc root-cause inspection of all 38 platform rework turns (`ANALYSIS_PLAN_v2.md` §5.3.4) showed that 22/38 were evaluator oracle false positives (affecting all arms equally, e.g., body sentences starting with "Table 1" flagged as missing `SEQ` caption fields), leaving only 16/780 (2.1%) true platform/model defects across 6 root causes, all of which were subsequently patched and covered by 148 unit tests.

6.2 Experiment 2: Pre-Write Citation and Two-Stage Neuro-Symbolic PHI Guards (`exp2b` & Locked `v2-phi` Test Split)

To quantify the complementary roles of Stage-1 deterministic validators (`scanPhi`/`redactPhi`) and the Stage-2 local small language model (`qwen2.5:1.5b` via loopback Ollama, `redactPhiCascaded`), we first evaluated Heurion 2.0 across 660 template test records (`exp2b`: 100 citation-guard sentences, 200 synthetic positive clinical notes containing 720 PHI spans across 7 categories, and 360 negative controls across 4 strata; Table 5), and then conducted a pre-specified, single-run evaluation on a locked, multi-specialty Chinese clinical de-identification test set (`v2-phi`, Table 6):

1. **Unregistered Inline Citations ($N = 100$ distinct sentences):** 40 bare DOIs across 8 biomedical publisher prefixes, 30 `https://doi.org/` URLs, and 30 bare PMIDs. All 100/100 (100%) were intercepted prior to CRDT mutation with `citation_not_registered`.
2. **Template Verification Benchmark (`exp2b`, 720 Spans in 200 Notes + 360 Negative Controls):** As shown in Table 5, **Stage 1 alone** (0.011 ms/note) achieved 100.0% (600/600) recall across the five structured and labeled categories, 0/200 false positives on numeric-dense PMC biostatistical paragraphs, and 0/100 false positives on invalid-checksum 16–18 digit serials, while **Stage 2 (`qwen2.5:1.5b` + post-validator)** captured 110/120 (91.7%) of template narrative names and addresses (710/720 = 98.6% combined recall) with 1/60 (1.7% combined across the two 30-sentence strata: 1/30 template and 0/30 independent) false positives on clean Chinese clinical sentences (vs. 60/60 without post-validation).
3. **Locked Multi-Specialty Clinical De-Identification Test Set (`v2-phi`, 100 Notes / 745 Spans + 200 Hard-Negative Sentences):** Because `exp2b`'s narrative clauses shared a single phrasing template, we constructed an independent benchmark (`ANALYSIS_PLAN_v2.md` §3) generated by an external LLM across diverse clinical departments and note types (admission notes, ward-round records, discharge summaries, family consultations) where 65.1% (485/745) of PHI spans occur in unlabeled free-form narrative prose and 34.9% (260/745) occur in labeled fields, paired with 200 hard-negative Chinese clinical sentences dense with eponyms, anatomical terms, drug regimens, and statistical results. After refining Stage-1 surname-prefix stripping and Stage-2 biomedical exclusion lists on the `dev` split (40 notes + 80 negatives), we executed a single locked run on the held-out `test` split (100 notes + 200 negatives; `TEST_RUN.lock`, Table 6) across four pre-specified ablation configurations:

   - `A_stage1` (*Stage-1 regex/checksum only*): Achieved 100.0% (209/209) span precision and 0.0% (0/200) hard-negative FPR, but recalled only 27.4% (204/745) of total PHI—capturing 71.9% (187/260) of labeled spans but only 3.5% (17/485) of narrative spans (0/360 narrative person names, 0/99 narrative addresses).
   - `B_platform` (*Stage 1 + Local 1.5B SLM + Deterministic Post-Validator*): Raised overall PHI recall by +59.6 percentage points to **87.0**% (**648/745**, Wilson 95% CI 84.4–89.2%), boosting narrative PHI recall from 3.5% to **83.3**% (**404/485**, CI 79.7–86.4%; narrative person names 289/360 = 80.3%, narrative addresses 92/99 = 92.9%, narrative workplaces 6/7 = 85.7%) and labeled recall to **93.8**% (**244/260**), while maintaining **97.6**% (**482/494**, CI 95.8–98.6%) span precision and a **6.0**% (**12/200**, CI 3.5–10.2%) hard-negative sentence FPR.
   - `C_slm_unvalidated` (*Stage 1 + Local 1.5B SLM without Post-Validator*): Recalled 87.7% (653/745) of PHI (only +5 spans over `B_platform`), but suffered catastrophic over-redaction of biomedical terminology: span precision collapsed to 56.2% (487/866) and hard-negative sentence FPR surged to **92.0**% (**184/200**).
   - `D_rules_only` (*Stage 1 + Broad Surname/Spatial Rule Candidates + Same Post-Validator, No SLM*): Recalled 98.8% (736/745) of PHI, but without the SLM's

Table 5: Heurion 2.0 pre-write citation and two-stage synthetic PHI guard benchmark (`exp2b`) comparing Stage-1 deterministic rules (0.011 ms/note) and Stage-2 local `qwen2.5:1.5b` SLM fallback across 720 synthetic positive PHI spans (200 notes) and 360 negative controls. Stage-2 false-positive cells for the clean Chinese sets report the model without post-validation; the combined column includes the post-validator.

Guard Subsystem / Evaluation Stratum	**Validation Mechanism**	**Test Inputs**	**Stage-1 Only (`scanPhi`)**	**Stage-2 Local SLM (`qwen2.5:1.5b`)**	**Combined Two-Stage Recall / FP Rate**
Bare DOI / URL / PMID Blocker	Pre-write AST regex	100 sents	100/100 (100%)	—	**100/100 (100.0%)**
Stale Block Overwrite Blocker	Block `base_rev` lock	50 trials	50/50 (100%)	—	**50/50 (100.0%)**
PHI: 18-Digit National ID (`id_card`)	ISO 7064 + date check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: 11-Digit Mobile (`phone`)	Prefix + boundary check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Medical Record (`medical_record_no`)	Clinical label match	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Bank / Settlement Card (`bank_card`)	ISO 7812 + Luhn mod-10	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Labeled Name + Diagnosis (`name_with_diagnosis`)	Alphanumeric/CJK subtype	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Unlabeled Narrative Name (`unlabeled_person_name`)	Local 1.5B SLM + surname guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI: Unlabeled Address / Org (`address_or_org`)	Local 1.5B SLM + spatial guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI Total Entity Recall (200 notes)	**Two-stage neuro-symbolic**	**720 spans**	600/720 (83.3%)	**+110/120 (91.7%)**	**710/720 (98.6%)**
Neg. Control: Real PMC Biostatistical Text	HR/CI/p-value/DOI text	200 paras	0/200 (0.0% FP)	skipped (no CJK; 0 calls)	**0/200 (0.0% FP)**
Hard Neg.: Invalid-Checksum 16–18 Digit IDs	ISO 7064 + Luhn vs. regex	100 serials	0/100 (0.0% FP)	—	**0/100 (0.0%)** vs. 100%
Hard Neg.: Clean Chinese Clinical Text (template set)	SLM ± post-validator	30 sents	0/30 (0.0% FP)	30/30 (unvalidated)	**1/30 (3.3% FP)**
Hard Neg.: Clean Chinese Clinical Text (independent set)	SLM ± post-validator	30 sents	0/30 (0.0% FP)	30/30 (unvalidated)	**0/30 (0.0% FP)**
Control: Rule-Only Stage-2 Candidates	Rules + same post-validator	120 spans / 60 sents	—	120/120 recall; 60/60 FP	—

Table 6: Locked test-split evaluation of the two-stage neuro-symbolic PHI funnel (`v2-phi`, `2026-10-22T15-08-56-203Z-test.json`; 100 diverse Chinese clinical notes containing 745 PHI spans [485 narrative, 260 labeled] and 200 hard-negative clinical sentences). Parentheses show Wilson 95% confidence intervals.

Ablation Configuration	**Overall Recall ($N = 745$ spans)**	**Narrative Recall ($n = 485$ spans)**	**Labeled Recall ($n = 260$ spans)**	**Span Precision (True / Total Pred.)**	**Hard-Negative FPR ($N = 200$ sents)**
`A_stage1` (Stage-1 Regex/Checksum Only)	27.4% (204/745) (24.3–30.7%)	3.5% (17/485) (2.2–5.5%)	71.9% (187/260) (66.2–77.0%)	**100.0%** (209/209) (98.2–100.0%)	**0.0%** (0/200) (0.0–1.9%)
`B_platform` (Stage 1 + 1.5B SLM + Validator)	**87.0% (648/745)** **(84.4–89.2%)**	**83.3% (404/485)** **(79.7–86.4%)**	**93.8% (244/260)** **(90.2–96.2%)**	**97.6% (482/494)** **(95.8–98.6%)**	**6.0% (12/200)** **(3.5–10.2%)**
`C_slm_unvalidated` (Stage 1 + 1.5B SLM, No Validator)	87.7% (653/745) (85.1–89.8%)	84.3% (409/485) (80.8–87.3%)	93.8% (244/260) (90.2–96.2%)	56.2% (487/866) (52.9–59.5%)	92.0% (184/200) (87.4–95.0%)
`D_rules_only` (Stage 1 + Broad Rules + Validator)	98.8% (736/745) (97.7–99.4%)	99.6% (483/485) (98.5–99.9%)	97.3% (253/260) (94.6–98.7%)	36.6% (567/1550) (34.2–39.0%)	93.5% (187/200) (89.2–96.2%)

contextual filtering, span precision dropped to 36.6% (567/1,550) and hard-negative FPR reached **93.5**% (**187**/**200**).

Design Implication: Local SLMs as Contextual Candidate Generators Between Deterministic Guards. Tables 5 and 6 demonstrate why neither deterministic rules nor a compact local SLM suffices alone for clinical de-identification. High-precision regex and checksum rules (`A_stage1`) miss 96.5% of narrative PHI, whereas broad candidate rules (`D_rules_only`) and unvalidated 1.5B SLM predictions (`C_slm_unvalidated`) corrupt over 92% of clean clinical sentences by misclassifying diseases, anatomical structures, and eponyms as person names or addresses. Sandwiching the local 1.5B SLM between Stage-1 pre-masking and a symbolic biomedical post-validator (`B_platform`) retains 99.2% (648/653) of the SLM's true PHI detections while eliminating 96.8% (360/372) of its false-positive span predictions, keeping all patient text within `127.0.0.1`.

6.3 Experiment 3: Longitudinal Ablation of Self-Evolving Procedural Memory (`exp3` & `v2-exp3`)

From v1 Exploration (1,079 Turns) to v2 Confirmatory Four-Arm Design (4,608 Turns). In our initial v1 longitudinal study (`ANALYSIS_PLAN.md`, 3 arms × 8 seeds × 5 sessions × 9 tasks = 1,079 analysed turns), most task archetypes overlapped with the 5 factory-seeded static rules, and newly crystallized rules for uncovered guard codes emitted only generic fallback advice ("call `doc_validate` before editing"). Consequently, while `utility` reduced total execution signals by 37% relative to `off` (`glmmTMB` RR = 0.63, Holm $p = 0.0011$; guard

Table 7: Confirmatory 4-arm longitudinal ablation of Heurion 2.0's self-evolving procedural memory (`v2-exp3`, `2026-10-22T17-11-50-247Z`, $N = 4,608$ analysed live LLM turns across 4 arms × 16 paired user seeds × 8 sessions [6 training + 2 held-out] × 9 uncovered constraint tasks). Primary analysis uses the three-way random-intercept Negative Binomial GLMM (`glmmTMB` in `stats.R`) with Holm multiplicity correction across H1 and H2; cluster-robust NB2 (`stats.py`), paired Wilcoxon signed-rank ($n = 16$ pairs), and paired bootstrap are reported as sensitivity analyses.

Experimental Arm	Users × Sessions	Analysed Turns	Guard Rejections / Turn (Total)	Held-Out (Sess. 7–8) Rejections / Turn	All Execution Signals / Turn	Task Completion Rate (All 8 Sess.)	Held-Out Completion
A. No Memory (`off`)	16 × 8	1,152	0.784 (903)	0.812 (234/288)	0.911 (1,050)	81.1% (934/1,152)	84.0% (242/288)
B. Static Seeded (`frozen`)	16 × 8	1,152	0.774 (892)	0.802 (231/288)	0.896 (1,032)	81.8% (942/1,152)	85.4% (246/288)
Pl. Placebo (`placebo`)	16 × 8	1,152	0.771 (888)	0.781 (225/288)	0.896 (1,030)	80.9% (932/1,152)	84.4% (243/288)
C. Self-Evolving (`utility`)	**16 × 8**	**1,152**	**0.285 (328)**	**0.194 (56/288)**	**0.321 (370)**	**92.8% (1,069/1,152)**	**100.0% (288/288)**
Pre-Specified Primary Mixed-Effects Models (`glmmTMB` NB2, `stats.R`, $N = 4,608$; Rate Ratios RR < 1 favor `utility`):							
H1 (`utility` vs. `off`) — Primary (Guard Rejections, `glmmTMB`):						RR = **0.363** (95% CI 0.320–0.412), $p = 1.32 \times 10^{-55}$, **Holm** $p = 2.64 \times 10^{-55}$	
H2 (`utility` vs. `frozen`) — Primary (Guard Rejections, `glmmTMB`):						RR = **0.368** (95% CI 0.324–0.417), $p = 3.87 \times 10^{-54}$, **Holm** $p = 3.87 \times 10^{-54}$	
Exploratory (`utility` vs. `placebo`) — Guard Rejections (`glmmTMB`):						RR = **0.369** (95% CI 0.325–0.419), $p = 1.32 \times 10^{-53}$	
Negative-Control Baselines (`frozen` vs. `off` / `placebo` vs. `off` / `frozen` vs. `placebo`):						RR = 0.988 ($p = 0.795$) / RR = 0.983 ($p = 0.723$) / RR = 1.005 ($p = 0.924$)	
Secondary Endpoints & Sensitivity Analyses:							
H1 & H2 Secondary (All Execution Signals, `glmmTMB` NB2):						H1 RR = **0.352** (**Holm** $p = 2.14 \times 10^{-66}$); H2 RR = **0.359** (**Holm** $p = 2.79 \times 10^{-64}$)	
Cluster-Robust NB2 by User (`stats.py`, $n = 16$ clusters):						H1 RR = 0.363 (95% CI 0.341–0.386); H2 RR = 0.368 (95% CI 0.345–0.392)	
Non-Parametric Paired User Tests ($n = 16$ seeds, `utility` lower in 16/16 users):						Wilcoxon $p = 4.34 \times 10^{-4}$ (vs. `off`) & 4.33×10^{-4} (vs. `frozen`); $\Delta_{\text{turn}} = -0.514$ & -0.479	
H3 (`utility` − `off`) — Completion Non-Inferiority & Superiority (Margin −5.0%):						**Δ** = **+11.72**% (95% CI [**+10.42**%, **+13.02**%] > −5.0%, **Non-Inferior & Superior**)	

rejections 0.039 vs. 0.078/turn, RR = 0.51, Holm $p = 0.1089$; cluster-robust NB2 RR = 0.50, Holm $p = 0.0269$), it did not significantly outperform static seeded rules `frozen` (0.039 vs. 0.053/turn, RR = 0.74, $p = 0.4263$).

To rigorously test whether online rule crystallization provides incremental value when agents encounter *novel, uncovered* collaborator and document constraints—and to separate rule content from static rules and prompt-length effects—we pre-specified a two-stage Phase-2 protocol (`ANALYSIS_PLAN_v2.md` §5–§6, frozen at tag `paper-eval-v2-p2`) over 9 authoring task archetypes (`TASKS_V2`: `open_comment_methods`, `open_comment_discussion`, `p_value_format`, `stat_symbol_replace`, `ambiguous_term_discussion`, `ambiguous_term_results`, `merge_list_items`, `table_merged_header`, and `out_of_scope_revert`) targeting 6 guard/revert codes *not* covered by the factory-seeded rules (`anchor_has_open_comments`, `text_not_found`, `ambiguous_match`, `invalid_structure`, `cell_not_found`, and `user_reverted_turn`). After verifying that our 432-turn 3-arm pilot (seeds 501–504, 4 users × 4 sessions × 9 tasks/arm) exceeded both pre-specified continuation thresholds (`off` rejection rate 0.785 ≥ 0.20/turn; `utility` vs. `frozen` RR = 0.43 ≤ 0.65, Holm $p = 1.25 \times 10^{-6}$, with held-out Session 4 rejections dropping to 0.056 vs. 0.778/turn), we executed the full 4,608-turn 4-arm confirmatory experiment (seeds 601–616):

- **Arm A — No Procedural Memory (`off`):** No procedural rules injected (1,152 turns).
- **Arm B — Static Seeded Rules Only (`frozen`):** Injects the 5 factory-seeded baseline rules ranked by context cues (1,000-char budget), with online rule crystallization disabled (1,152 turns).
- **Arm Pl — Length-Matched Placebo Control (`placebo`):** Same as `frozen`, plus a length-matched block of task-irrelevant platform background descriptions to control for prompt-length and formatting effects (1,152 turns).
- **Arm C — Self-Evolving Procedural Memory (`utility`):** Factory-seeded rules plus online signal crystallization (`LEARNED_RULE_TEMPLATES`, threshold 2 occurrences) and cue-ranked injection (1,152 turns).

Each arm ran 16 independent simulated user trajectories (seeds 601–616) across 8 consecutive sessions (6 training sessions on 7 training PMC manuscripts, followed by **Sessions 7 and 8 on 3 held-out PMC manuscripts**) × 9 shuffled tasks per session, totaling 4,608 turns (100% completed with 0 infrastructure or setup exclusions).

Table 7 and Figure 2 establish four findings from the 4,608-turn confirmatory experiment:

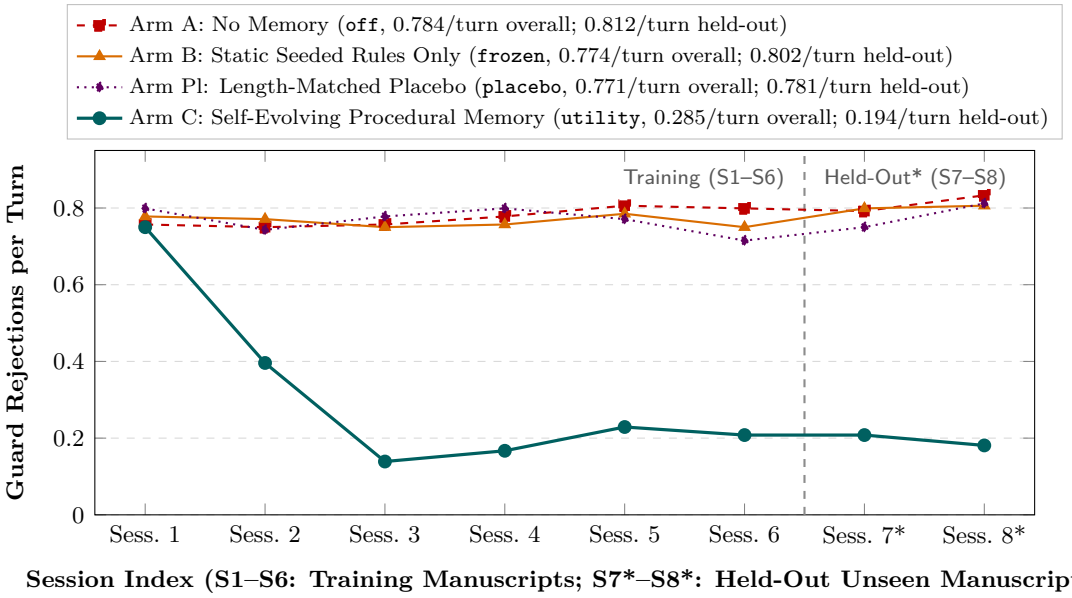


Figure 2: Longitudinal guard-rejection trajectories across 8 consecutive sessions (144 turns/session/arm, 4,608 total turns across 16 paired user seeds) in the confirmatory Phase-2 experiment (`v2-exp3`). All four arms start at $\approx$ 0.75–0.80 rejections/turn in Session 1. While `off`, `frozen`, and `placebo` remain flat across all 8 sessions ($\approx$ 0.72–0.83/turn), `utility` autonomously crystallizes all 6 uncovered rules during Sessions 1–2 (96 rules across 16 users), reducing guard rejections to 0.139–0.229/turn in Sessions 3–6 and generalizing zero-shot to unseen clinical manuscripts in held-out Sessions 7*–8* (0.208 and 0.181/turn, with 100% [288/288] task completion).

1. **Autonomous Crystallization of All Uncovered Constraints:** Every one of the 16 independent simulated users in Arm C (`utility`) autonomously crystallized all 6 uncovered procedural rules during Sessions 1–2 (96 learned rules total). Consequently, pre-write guard rejections fell from 0.750/turn (108 rejections) in Session 1 to 0.396/turn (57) in Session 2 and 0.139/turn (20) in Session 3, remaining low across subsequent training sessions (0.167–0.229/turn in Sessions 4–6).

2. **Causal Isolation Against Static Rules and Prompt-Length Placebo (H1 & H2):** Under the pre-specified negative binomial mixed model (`glmmTMB`), `utility` reduced guard rejections by 63.7% vs. `off` (RR = 0.363, 95% CI 0.320–0.412, Holm $p = 2.64 \times 10^{-55}$), by 63.2% vs. `frozen` (RR = 0.368, 95% CI 0.324–0.417, Holm $p = 3.87 \times 10^{-54}$), and by 63.1% vs. `placebo` (RR = 0.369, 95% CI 0.325–0.419, $p = 1.32 \times 10^{-53}$). Crucially, the three negative-control contrasts—`frozen` vs. `off` (RR = 0.988, $p = 0.795$), `placebo` vs. `off` (RR = 0.983, $p = 0.723$), and `frozen` vs. `placebo` (RR = 1.005, $p = 0.924$)—were statistically indistinguishable from 1.00, proving that neither unaligned static rules nor extra prompt formatting reduces errors on novel constraints. Non-parametric Wilcoxon tests confirmed that 16/16 (100%) users had lower rejection rates under `utility` than under `off`, `frozen`, or `placebo` ($p \leq 4.34 \times 10^{-4}$).

3. **Task Completion Superiority and Latency Reduction (H3):** Rather than avoiding guards by abstaining from edits, `utility` raised overall task completion from 81.1% (`off`), 81.8% (`frozen`), and 80.9% (`placebo`) to **92.8**% (**1,069/1,152**), a paired gain of +11.72% (95% Bootstrap CI [+10.42%, +13.02%], far exceeding the −5.0% non-inferiority margin), while reducing overall mean turn latency from 15.0–15.5 s in the control arms to 12.4 s (−17% to −20%).

Table 8: Automated verification and empirical evaluation coverage across the Heurion 2.0 codebase.

Verification Metric	**Measured Value**
Automated Test Suite (`apps/platform/tests/*.test.ts`)	74+ Files / 625+ Tests Passing (100%)
OOXML Word Fidelity Corpus (`v2/fidelity/run.ts`, 47 fixtures × 4 layers)	188/188 Checks Passing (100%)
Deterministic & Local-SLM Guard Records (`exp1`, `exp2b`, locked `v2-phi` test set)	1,010 Records (1,465 PHI Spans + 560 Neg. Controls)
Three-Arm Manuscript Co-Editing Live Turns (`exp2a-papers` + `v2-q1` 1a & 1b)	4,650 Turns (20 CC BY PMC Manuscripts)
Longitudinal Procedural Memory Live Turns (`exp3` v1 + `v2-exp3` Pilot & 4-Arm Formal)	6,119 Turns (4,608 in 4-Arm Confirmatory Study)
Total Analysed Live LLM Evaluation Turns	**10,769 Turns**

4. **Zero-Shot Generalization to Held-Out Manuscripts and Turn-Revert Recovery:** In held-out Sessions 7–8 on 3 unseen PMC manuscripts, `utility` reduced guard rejections by 76.1% relative to `off` (56/288 = 0.194 vs. 234/288 = 0.812/turn), achieved **100.0**% (**288**/**288**) task completion, and lowered mean turn latency to 11.35 s (vs. 15.0–15.8 s in controls, −25%). Furthermore, on `out_of_scope_revert` (128 turns/arm), where clinicians revert turns that modify adjacent paragraphs beyond the requested block, `off`, `frozen`, and `placebo` succeeded in only 6/128 (4.7%), 3/128 (2.3%), and 10/128 (7.8%) turns, whereas `utility` experienced 2 reverts per user in Sessions 1–2 (32 total), crystallized `user_reverted_turn`, and succeeded in **96**/**96** (**100.0**%) turns across Sessions 3–8.

6.4 Pre-Specification Transparency, Deviations, and Limitations

Pre-Specification and Git Provenance. Both generations of the evaluation protocol (`paper/benchmarks/ANALYSIS_PLAN.md` and `paper/benchmarks/v2/ANALYSIS_PLAN_v2.md`) were frozen in the git repository prior to formal data collection at tags `paper-eval-v1` (`1d8f307c`), `paper-eval-v1.1` (`e060107f`), `paper-eval-v2-p1a` (`16180f3c`), `paper-eval-v2-p1b` (`932f4e3a`), and `paper-eval-v2-p2` (`e8362c88`). Because protocols were locked via internal git tags rather than an external registry such as OSF, we refer to them as pre-specified rather than externally pre-registered. All developmental iterations (such as the Phase 1a-to-1b `F1`–`F5` platform fixes, the 38-turn post-hoc attribution in §5.3.4, and the Phase-2 pilot continuation gate in §5.2) are recorded chronologically in `ANALYSIS_PLAN_v2.md` and reported alongside confirmatory results.

Limitations. Our empirical evaluation has five main limitations: (1) all live agent authoring experiments used `deepseek-flash`, while Stage-2 local PHI extraction used `qwen2.5:1.5b`; (2) all user trajectories (16 seeds in `v2-exp3`, 20 PMC manuscripts across `v2-q1` and `v2-exp3`) were driven by automated scripts with deterministic clinician callbacks rather than prospective human-subject user trials (a clinician crossover study protocol is outlined in `ANALYSIS_PLAN_v2.md` §7 for future work); (3) although the 100-note `v2-phi` test set substantially expands narrative variety (485 narrative spans across multiple clinical specialties) over the initial template set, external validation on annotated multi-center hospital EHRs [10] remains necessary before clinical deployment; (4) while citation support is checked at insertion time via abstract matching and evaluated by an LLM judge in `v2-q1`, blind human specialist grading of `verify_claims` verdicts is left to future work; and (5) task completion in `exp3` is graded by deterministic structural/textual oracles, with blind human audit of drawn subsamples (`audit/items.json`) deferred to future work.

Platform Verification Summary. Table 8 summarizes the automated verification and empirical evaluation coverage across the repository.

7 Related Work

Collaborative Editing Data Structures and Agent-Computer Interfaces (ACI). Conflict-free Replicated Data Types (CRDTs) [16] and local-first software architectures [17] enable concurrent multi-actor editing without central locking. In parallel, SWE-agent [3] and OpenHands [4] showed that specialized Agent-Computer Interfaces outperform raw shell scripting for software engineering, and the Model Context Protocol (MCP) [18] standardized tool exposure for LLM agents. Heurion 2.0 combines these threads by exposing a shared ProseMirror + Yjs CRDT document core through guarded MCP primitives with node-level optimistic concurrency locks, per-turn selective CRDT undo, and raw-slice OOXML export.

Literature Synthesis, Citation Hallucination, and De-Identification. Empirical audits have documented frequent fabricated references in LLM-generated medical content [9] and in LLM-generated bibliographies across disciplines [8]; see [7] for a broader survey of hallucination. While STORM [5] and PaperQA2 [6] automate literature synthesis and FActScore [20] and ALCE [21] evaluate post-hoc attribution, Heurion 2.0 enforces citation registration as a pre-write syntactic contract (`[@c:id]`). For clinical privacy, rule-based and hybrid de-identification systems such as the i2b2/UTHealth shared-task systems [10], Philter [11], and Microsoft Presidio [19] combine regular expressions with statistical or neural sequence labelers; our two-stage PHI funnel adapts this hybrid pattern to local loopback deployment by pairing checksum/subtype validators with a pre-masked 1.5B SLM and symbolic output validation.

Self-Evolving Agents and Memory Architectures. Reflexion [14] and MemGPT [15] introduced verbal reinforcement and tiered memory management for LLM agents, while Voyager [22] accumulated executable skills in open-ended environments. In collaborative authoring, mixing subjective user stylistic preferences with operational error recovery can create prompt interference; Heurion 2.0 separates human-confirmed declarative preferences from signal-crystallized procedural rules ranked by document-state and prompt cues.

8 Conclusion

We have presented Heurion 2.0, a collaborative clinical authoring platform built around a contract-guarded dual-surface CRDT engine, a local two-stage neuro-symbolic PHI funnel, and a self-evolving procedural memory system:

- **Dual-Surface CRDT Co-Editing and OOXML Export:** Replacing direct `python-docx` file scripting with a shared Yjs + ProseMirror CRDT core, stable block IDs (`{#id}`), `base_rev` concurrency guards, per-turn selective undo (`revertTurn`), and raw XML slice caching (`node_src`) reduced manuscript rework rates to 4.9% (38/780; 2.1% true defect rate, 0 lost human edits) vs. 14.0% (OR = 32.47, Holm $p = 1.03 \times 10^{-16}$) for direct file scripting and 10.3% (OR = 9.66, Holm $p = 1.23 \times 10^{-8}$) for tool-assisted file scripting across 2,340 confirmatory turns on 10 unseen 2026 PMC manuscripts, while lowering API cost by 30%–33% (GMR = 0.70 and 0.67, Holm $p < 10^{-72}$).
- **Pre-Write Citation and Local Two-Stage PHI Guards:** Pre-write syntactic validation blocked 100/100 unregistered inline DOIs/PMIDs, and pairing Stage-1 regex/checksum rules with a pre-masked local 1.5B SLM (`qwen2.5:1.5b`) and symbolic biomedical output validation recalled 87.0% (648/745) of PHI spans (83.3% [404/485] on narrative spans and 93.8% [244/260] on labeled spans) at 97.6% (482/494) span precision and 6.0% (12/200) hard-negative sentence FPR on the locked multi-specialty clinical test set, outperforming both Stage-1 rules alone (27.4% recall, 3.5% narrative) and the unvalidated SLM (92.0% hard-negative FPR).

- **Self-Evolving Procedural Memory:** In our confirmatory 4,608-turn, 4-arm (`off`, `frozen`, `placebo`, `utility`) × 16-user × 8-session longitudinal experiment on uncovered structural and collaborator constraints, all 16 `utility` users autonomously crystallized all 6 uncovered rules (96 rules total), reducing pre-write guard rejections per turn by 63.7% overall (0.285 vs. 0.784 [`off`], 0.774 [`frozen`], and 0.771 [`placebo`]; `glmmTMB` RR = 0.363–0.369, Holm $p < 10^{-53}$) and by 76.1% on held-out unseen manuscripts (0.194 vs. 0.812/turn), while raising task completion by +11.72% (92.8% vs. 81.1%; 100.0% [288/288] in held-out sessions).

Statements and Declarations

Ethics Statement. No human subjects were recruited and no real patient electronic health records or identifiable clinical data were accessed in this study. All 20 clinical manuscripts used across Experiments 1b, 1c, and 3 are open-access articles distributed under Creative Commons (CC BY) licenses via PubMed Central, all PHI evaluation strings in Experiment 2 (`exp2b` and `v2-phi`) were synthetically generated, and all user trajectories were executed by automated seeded scripts.

Code and Data Availability. The platform source code (`apps/platform/`), benchmark harnesses (`paper/benchmarks/`), pre-specified analysis plans (`paper/benchmarks/ANALYSIS_PLAN.md` and `paper/benchmarks/v2/ANALYSIS_PLAN_v2.md`, git tags `paper-eval-v1`, `paper-eval-v1.1`, `paper-eval-v2-p1a`, `paper-eval-v2-p1b`, and `paper-eval-v2-p2`), raw turn-level JSONL/CSV logs, and statistical analysis scripts (`stats.R`, `stats.py`) are included in the project repository workspace (`paper/results/`).

Competing Interests. Hui Zhao is affiliated with Heurion Research Lab, which develops the Heurion 2.0 platform evaluated in this paper. Dong Qian declares no competing financial interests.

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